

Pablo M. Sánchez Alarcón

SPACE MISSIONS · DATA SCIENCE · GALAXY EVOLUTION

NPP fellow at NASA Ames, California

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Summary

Astrophysicist working at the interface of **low-surface-brightness (LSB) science requirements, calibration and systematics, and pipeline development** for space telescope missions. My experience spans the full mission lifecycle, from proposal design (PI, HST Cycle 32) through commissioning and calibration (*Roman* zodiacal light, stray light, and PSF characterization) to pipeline delivery (ROSALIA, astropipe, ELIPSE). My work has enabled LSB science for the wider community, from reference catalogs used by teams preparing for *Euclid*, Rubin, and *Roman*, to surface-brightness software integrated into ESA's Datalabs platform. Four first-author and 12 co-authored publications on LSB galaxy structure using *Spitzer*, *Hubble*, *Euclid*, and *Roman* data, and over 8 years building Python-based science software adopted by mission teams beyond my own research. Currently working as a NASA Postdoctoral Fellow at NASA Ames, California, working on subtraction of galactic cirrus on Roman Space Telescope using machine learning, and part of the team developing the ROSALIA pipeline.

Research Interests:

- 👁 **Low Surface Brightness Universe:** Dwarf galaxies, UDGs, extended discs, tidal structures, and diffuse emission preservation.
- 👁 **Observational Astronomy:** Observation planning, image calibration, and data reduction across ground- and space-based facilities.
- 👁 **Space Mission Calibration:** Stray light, zodiacal light, extended PSF, background systematics, and commissioning support.
- 👁 **Pipeline & Software Development:** Python science pipelines, ML/AI for systematics removal, GPU-parallel processing.

Professional Experience

NASA Ames Research Center

California, USA

POSTDOCTORAL FELLOW

Jan. 2026 - today

- Selected for the **NASA Postdoctoral Program** fellowship (NPP) to lead research on **Low Surface Brightness (LSB)** science using data from ***Euclid*** and ***Roman*** space telescopes. Developing novel **deep learning** frameworks to detect and characterize faint galactic structures and **galactic cirrus** in deep Roman space imaging.
- Contributing developer on **ROSALIA** (Roman Sky Analyst for Low Surface Brightness Imaging & Astronomy), the mission's LSB sky-analysis pipeline.
- Supporting Roman commissioning activities: zodiacal light, stray light, and extended PSF model calibration to assess LSB capabilities.

Instituto de Astrofísica de Canarias

Tenerife, Spain

PREDOCTORAL FELLOW

Sept. 2021 - Sept. 2025

- Conducted astrophysical research toward completion of the PhD, specializing in observational astronomy and low-surface-brightness.
- Analysis of state-of-the-art deep imaging from different ground- and space-based telescopes (HST, *Spitzer*, *Euclid*, INT, VLT) and surveys (Legacy Survey, SDSS, DES). Studying the faint outskirts of galaxies, with emphasis on developing software optimized for low-surface-brightness (LSB) science. Experience reducing and calibrating images with techniques optimized for LSB. Software development of end-to-end pipelines.
- Supporting observational campaigns (over 50 nights in professional observatories), from observational planning of surveys to conducting observations to reducing the data.

Observatoire Astronomique de Strasbourg

Strasbourg, France

RESEARCH VISITOR

March - May 2025

- Collaborating with Dr. Pierre-Alain Duc to develop low-surface-brightness techniques for *Euclid* imaging and analysis.

University of Oulu

Oulu, Finland

RESEARCH VISITOR

March - June 2023, December 2024

- Research collaboration with Dr. Heikki Salo: calibration and analysis of *Spitzer* images for the S⁴G survey.

Astronomical Observatory of the U. Autónoma de Madrid

Madrid, Spain

INTERNSHIP FOR DATA ANALYSIS

Sept. 2018 - June 2019

- Developed a Python-based data reduction and calibration pipeline for the observatory's CMOS imager while operating as lead telescope operator, supporting instrument maintenance, and delivering calibrated survey data used by degree students — early experience spanning software development, instrument operation, and end-user data delivery.

Education

Universidad de la Laguna

Tenerife, Spain

PH.D. IN ASTROPHYSICS

GALAXY EVOLUTION AND DISC STRUCTURE THROUGH DEEP SURFACE PHOTOMETRY.

Supervisor: Prof. Johan Knapen. **Co-supervisors:** Dr. Javier Román and Dr. Sébastien Comerón.

Sept. 2021 - August 2025

- Investigated disc structure evolution from isolated galaxies to *Euclid*'s statistical sample of thousands of discs out to $z \sim 1$, and built the Complete *Spitzer* Survey of Stellar Structure in Galaxies (CS⁴G), a 3,239-galaxy reference catalog for *Euclid*, Rubin, and Roman.
- As a part of my thesis, I participated in several training activities, such as doctoral schools, congress, observational campaigns, research stays, and seminars. I have also been on the IAC's seminars committee for three years, and participated in outreach events.
- Grant:** FPI-2020
- Defence Date:** 29 of July 2025. Graduated with the highest mark, **Cum Laude** distinction, and international mention.

Universidad Autónoma de Madrid & Instituto de Física Teórica

Madrid, Spain

M.Sc. IN THEORETICAL PHYSICS, SPECIALIZATION IN ASTROPHYSICS AND PHYSICS OF THE COSMOS, TOTAL 60 ECTS

Sept. 2019 - July. 2020

- **M.Sc. Thesis (with honours) FABADA: Fully Adaptive Bayesian Algorithm for Data Analysis:** Novel noise reduction method, specifically for astronomical data in 1D and 2D. Implemented in **Python** and published in **RASTI**.
- **Grant:** Beca Fondo Social U. Autónoma de Madrid

Universidad Autónoma de Madrid

Madrid, Spain

B.Sc. IN PHYSICS, TOTAL 240 ECTS

Sept. 2014 - July. 2019

- **Grant:** Ministerio de Educación y Formación Profesional (MEC)

Accepted Proposals as PI

Hubble Space Telescope (Cycle 32): 2 orbits with ACS@WFC (Scheduled Feb 2026). Awarded a \$53k grant. [\[Proposal Link\]](#)

William Herschel Telescope (2025B): 3 hours (3 observing blocks) with the WEAVE LIFU instrument.

Isaac Newton Telescope (2024A): 5 nights with the Wide Field Camera (WFC).

William Herschel Telescope (2023B): 5 hours (5 observing blocks) with the WEAVE LIFU instrument.

SARA Telescope Network (2023B): 46 hours across 1 m telescopes (7 nights at Cerro Tololo, Chile; 3 nights at JKT, La Palma).

I have also participated as Co-I on other accepted proposals, including [The Roman eXtreme Deep Field \(RXDF\)](#).

Observational Experience

Over 50 nights of professional observing experience across six facilities, spanning optical imaging and spectroscopy:

35 nights **Isaac Newton Telescope**, 2.54 m, Wide Field Camera, Observatorio Roque de los Muchachos

La Palma, Spain

7 nights **SARA Telescope**, 1.0 m, Cerro Tololo Inter-American Observatory

Cerro Tololo, Chile

4 nights **CAHA Telescope**, 2.2 m, CAFOS instrument, Calar Alto Observatory

Almería, Spain

3 nights **Jacobus Kapteyn Telescope**, 1.0 m, Observatorio Roque de los Muchachos

La Palma, Spain

2 nights **Carlos Sánchez Telescope**, 1.52 m, MusCAT2 instrument, Teide Observatory

Tenerife, Spain

6 hours **Green Bank Telescope**, 100 m, VEGAS instrument, Green Bank Observatory

West Virginia, USA

Software

ROSALIA Core contributor to Roman Space Telescope's low-surface-brightness sky-analysis pipeline.

FABADA Author of open-source Bayesian noise-reduction package (published RASTI, 2023). ★ 25 📄 3

ELIPSE Developer of end-to-end automated pipeline for galaxy surface photometry using *Euclid* data.

astropipe Creator of an astronomical toolkit for extended source masking, structural fitting, and surface photometry.

FORECAST Forward-models cosmological simulations into realistic synthetic astronomical images. Contributor for optimization.

Scientific Meetings

EAS2026

Lausanne, Switzerland

E-POSTER & CONTRIBUTED TALK: "ROSALIA: ENABLING ULTRA-LOW SURFACE BRIGHTNESS IMAGING WITH ROMAN"

July 2026

Working with Euclid Q1: From Data Access to Science

Madrid, Spain

CONTRIBUTED TALK: "DISC BREAKS HIGHLIGHT THE STRUCTURAL EVOLUTION OF GALAXIES THROUGH COSMIC TIME"

Dec. 2025

Galaxy Memoirs: Inferring their past from their present

Búzios, Brazil

CONTRIBUTED TALK: "UNVEILING DISK BREAK EVOLUTION IN GALAXIES WITH EUCLID"

Aug. 2025

Euclid from Q1 to DR1 (Joint Science Working Groups Meeting)

Tenerife, Spain

CONTRIBUTED TALK: "EVOLUTION OF DISC BREAKS AS SEEN BY EUCLID"

Jan. 2025

LSST@Europe 6

La Palma, Spain

LOC MEMBER & CONTRIBUTED TALK: "CS⁴G: DISC GALAXY EXTENSION AND SURFACE BRIGHTNESS PROFILES"

Sep. 2024

XXXIIst IAU General Assembly

Cape Town, South Africa

CONTRIBUTED TALK: "CS⁴G: DISC GALAXY EXTENSION AND SURFACE BRIGHTNESS PROFILES" [ID: 2469]

Aug. 2024

3rd Annual Meeting of the CAVITY Project

Granada, Spain

CONTRIBUTED TALK: "INT DATA CAMPAIGNS AND SURFACE BRIGHTNESS PROFILES"

Jun. 2023

XV Reunión Científica de la Sociedad Española de Astronomía (SEA)

Tenerife, Spain

E-POSTER & CONTRIBUTED TALK: "FULLY ADAPTIVE BAYESIAN ALGORITHM FOR DATA ANALYSIS (FABADA)"

Sep. 2022

XXXIst IAU General Assembly

Busan, Korea

TWO CONTRIBUTED E-POSTERS (ONLINE)

Aug. 2022

European Astronomical Society (EAS) Annual Meeting

Valencia, Spain

TWO CONTRIBUTED E-POSTERS: AMIGA & FABADA

Jun. 2022

Schools and Courses

Oct. 2024	Accurate Flux Calibration in the Era of Space Astronomy & All-Sky Surveys , STScI	Online
Jun. 2022	Summer School in Astroinformatics II , Penn State Univ.	Online
Jun. 2022	Summer School in Astrostatistics , Penn State Univ.	Online
May 2022	IAA-CSIC Severo Ochoa Advanced School on Galaxy Evolution , IAA	Granada, Spain
Feb. 2022	DAWN Winter School , Cosmic Dawn Center	Online
Nov. 2021	2nd IAA-CSIC Severo Ochoa School on Statistics, Data Mining, & ML , IAA	Granada, Spain
Nov. 2021	IAA Severo Ochoa Advanced School on Star Formation , IAA	Granada, Spain
Sep. 2020	Machine Learning Specialization , Univ. of Washington	Online
Feb. 2020	XV SVO School , Spanish Virtual Observatory	Madrid, Spain

Scientific Activities

INVITED TALKS

2025	Invited Talk , Observatório do Valongo	Rio, Brazil
2025	Invited Lecture , Automated Image Analysis at the “EDUCADO” Observing School	La Palma, Spain
2025	Invited Seminar , Strasbourg Observatory	Strasbourg, France
2025	Invited Talk , “EDUCADO” Mid-Term Meeting	La Palma, Spain
2024	Invited Talk , University of the Witwatersrand	Johannesburg, S.A.
2023	Invited Talk , Celebrating John Beckman’s 60 Years in Astrophysics (IAC)	Tenerife, Spain
2023	Invited Seminar , University of Oulu	Oulu, Finland

ACADEMIC SERVICES

2026	Referee , Monthly Notices of the Royal Astronomical Society	MNRAS
2024	Referee , Astrophysics & Space Science Journal	SpringerNature
2022–24	Committee Member , Seminars Committee at the IAC	IAC

TEACHING & SUPERVISION

2026	M.Sc. Co-Supervisor , Matías Salah: Galaxy halos in dwarf galaxies using deep GTC imaging	VIU
2026	M.Sc. Supervisor , Marta Sánchez: ML classification of galaxy surface brightness profiles (<u>Honors & 2k€ Prize</u>)	UCM
2022	B.Sc. Co-Supervisor , Carlos Gil-Igual: PSF deconvolution & noise reduction (Adaptive Wiener Filter)	UAM
2022	B.Sc. Co-Supervisor , Pablo Jimenez Sánchez: Sky background subtraction & signal detection	UAM

OUTREACH ACTIVITIES

2024	Invited Speaker , “Amigos del IAC” Public Seminar [Video]	Online
2019	Workshop Lead , Astrophotography Workshop (3 sessions), IV Jornadas Astronómicas	UAM, Madrid
2018–19	Observational Support , Guided >20 public observation nights at the UAM Observatory	UAM, Madrid
2015	Invited Speaker , Astrobiology Talk, “Universal Fun Facts”, Biology Dept.	UAM, Madrid

Skills

PROGRAMMING LANGUAGES

Advanced Skills	Python, Linux, SQL, Shell
Basic Skills	C/C++, R, Julia, Matlab/Octave, JavaScript.

SOFTWARE

Advanced Skills	LaTeX, SExtractor, SWarp, SCamp, PSFEx, GNUastro, Topcat, Astropy, DATALABS, romanisim,
Intermediate Skills	Streamlit, OpenMP, CUDA, Numba, ALADIN, ESASky, IRAF, romancom
Basic Skills	MPI, MongoDB, TensorFlow, STILTS, Grafana, TuriCreate, Poetry

LANGUAGES

Native tongue	Spanish
Advanced	English
Basic	French, learning

Refereed Publications

I have published 4 first-author papers in refereed journals, co-authored 12 articles, and published one proceeding. A completed list of all publications I have been involved can be found in this [ADS Library](#) 📖

AS FIRST AUTHOR:

- 👁 **Sánchez-Alarcón, P. M.**, Román, J., Comeron, S., Knapen, J. H., Huertas-Company, M., et al. (2026). Euclid Quick Data Release (Q1): Disc breaks highlight the structural evolution of galaxies through time. Submitted to A&A. DOI: [10.48550/arXiv.2609.26877](#).
- 👁 **Sánchez-Alarcón, P. M.**, Salo, H., Knapen, J. H., Comerón, S., Román, J., Watkins, A. E., Buta, R. J., Laine, S., Falcón-Ramrez, J. M., Anetjärvi, M., Athanassoula, E., Bosma, A., Gadotti, D. A., Hinz, J. L., Ho, L. C., Holwerda, B. W., Janz, J., Kim, T., Koda, J., . . . Zaritsky, D. (2025). The Complete Spitzer Survey of Stellar Structure in Galaxies (CS4G). A&A 697 A38. [10.1051/0004-6361/202451641](#).
- 👁 **Sánchez-Alarcón, P. M.**, Román, J., Knapen, J. H., Verdes-Montenegro, L., Comerón, S., Rich, R. M., Beckman, J. E., Argudo-Fernández, M., Ramírez-Moreta, P., Blasco, J., Unda-Sanzana, E., Garrido, J., & Sánchez-Exposito, S. (2023). The AMIGA sample of isolated galaxies. XIV. Disc breaks and interactions through ultra-deep optical imaging. A&A, 677. DOI: [10.1051/0004-6361/202346719](#)
- 👁 **Sánchez-Alarcón, P. M.**, & Ascasibar, Y. (2023b). Fully adaptive Bayesian algorithm for data analysis: FABADA. RAS Techniques and Instruments, 2 (1). DOI: [10.1093/rasti/rzad006](#)

AS CO-AUTHOR:

- 👁 Román, J., Rich, R. M., **Sánchez-Alarcón, P. M.**, Jiménez-Teja, Y., Li, C., & Esquide, M. (2026). The HERON Coma Project: Exploring the low-surface-brightness frontiers in the Coma cluster. Forthcoming article in A&A. DOI: [10.1051/0004-6361/202662310](#)
Contribution: Led the analysis of the surface brightness profiles. Writing and revision of the manuscript.
- 👁 Borlaff, A. S., Marcum, P. M., Howell, S. B., **Sánchez-Alarcón, P. M.**, Dubois, D., & McDowell, J. (2026). SPHEREx Confirms Predictions for Artificial Satellite Trail Pollution in Low Earth Orbit. AJ, 172, 41. DOI: [10.3847/1538-3881/ae71ce](#)
Contribution: Contributed to visual satellite trail detection and analysis. Writing and revision of the manuscript.
- 👁 Annibali, F., Ferguson, A. M. N., **Sánchez-Alarcón, P. M.**, et al. (2026). Euclid: Early Release Observations – The extended stellar component of the IC10 dwarf galaxy. A&A, 712, A175. DOI: [10.1051/0004-6361/202658966](#)
Contribution: Surface brightness analysis of the data, leading the morphological decomposition, contributing to the writing and revision of the manuscript.
- 👁 Urbano, M., Duc, P.-A., Poulain, M., (including **Sánchez-Alarcón, P. M.**), et al. (2026). Euclid preparation LXXIX. Using mock low surface brightness dwarf galaxies to probe Euclid Wide Survey detection capabilities. A&A, 707, A229. DOI: [10.1051/0004-6361/202557270](#)
Contribution: Analysis of the surface photometry and techniques: contributing to the revision of the manuscript.
- 👁 Mondelin, M., Bournaud, F., Cuillandre, J. C., (including **Sánchez-Alarcón, P. M.**) et al. (2025). *Euclid*: Early Release Observations – The surface brightness and colour profiles of the far outskirts of galaxies in the Perseus cluster. A&A, 689, A213. DOI: [10.1051/0004-6361/202554838](#)
Contribution: Member of the consortium. Contributing to the discussion of the analysis, techniques used, and review of the manuscript.
- 👁 Perez, I., Gil, L., Ferre-Mateu, A., Torres-Rios, G., (including **Sánchez-Alarcón, P. M.**) et al., (2025). Galaxy mass-size segregation in the cosmic web from the CAVITY parent sample. A&A, 695, A84. DOI: [10.1051/0004-6361/202452514](#)
Contribution: Helping with the analysis of the results and suggesting the methods and techniques to follow. Interpreting the findings from my expertise in disc outskirts. Contributing to the review of the manuscript.
- 👁 Garcia-Benito, R., Jiménez, A., Sánchez-Menguiano, L., Ruiz-Lara, T., (including **Sánchez-Alarcón, P. M.**) et al. (2024). CAVITY: Calar Alto Void Integral-field Treasury survey: I. First public data release. A&A, 691, A161. DOI: [10.1051/0004-6361/202451400](#)
Contribution: Helping with the observations of the data. Contributing to the review of the manuscript.
- 👁 McCall, C., Jermak, H. E., Steele, I. A., Kobayashi, S., Knapen, J. H., & **Sánchez-Alarcón, P. M.** (2024). Detection of an intranight optical hard lag with colour variability in blazar PKS 0735+178. MNRAS, 528 (3). DOI: [10.1093/mnras/stae310](#)
Contribution: Helping with the observations of the data. Contributing to the reduction of the data and review of the manuscript.
- 👁 Pérez, I., Verley, S., Sánchez-Menguiano, L., Ruiz-Lara, T., (including **Sánchez-Alarcón, P. M.**) et al. (2024). CAVITY, Calar Alto Void Integral-field Treasury survey and project extension. A&A, 689, A213. DOI: [10.1051/0004-6361/202449749](#)
Contribution: I am an active member of the collaboration. I conducted over 30 observation nights for the deep imaging data. I contributed by analysing deep images, interpreting findings, and writing and reviewing the manuscript.
- 👁 Sánchez, S. F., Garcia-Benito, R., González Delgado, R., (including **Sánchez-Alarcón, P. M.**) et al. (2024). The CAVITY Project: Spatially-Resolved and Characteristic Properties of Galaxies Derived Using pyPipe3D. *Rev. Mexicana Astron. Astrofis.*, 60, 323–341. DOI: [2024RMxAA..60..323S](#)
Contribution: Contributing to review the manuscript. Part of the survey team and followed the analysis.
- 👁 Román, J., Rich, R. M., Ahvazi, N., Sales, L. (including **Sánchez-Alarcón, P. M.**) et al. (2023). A giant thin stellar stream in the Coma Galaxy Cluster. A&A, 679, A157. DOI: [10.1051/0004-6361/202346780](#)
Contribution: Contributing to the analysis of the data. Discussion of methods, techniques and findings. Contributing to the review of the manuscript.
- 👁 Román, J., **Sánchez-Alarcón, P. M.**, Knapen, J. H., & Peletier, R. (2023). Evidence for globular cluster collapse after a dwarf-dwarf merger: A potential nuclear star cluster in formation. A&A, 671, L7. DOI: [10.1051/0004-6361/202345928](#)
Contribution: Helping with the analysis of the data and interpreting the findings. Contributing to writing and editing the manuscript. This research led to the HST and WEAVE observation proposal being accepted as PI.